

Laboratory Data Analysis and Presentation

Science

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Laboratory Data Analysis and Presentation (A07884)

Short Title: Lab Data Analysis & Pres
Faculty Science and Computing
Department Science
Cluster IT
Author KGRENNAN
Credits 5

Level: Introductory

Description of Module / Aims

In this module students will develop skills in the use of spreadsheets to visualise, plot, analyse and interpret laboratory data effectively. A focus will be on the use of linear regression analysis to analyse laboratory data from a range of experiments in their discipline. An introduction into the use of chemical drawing software will also be provided, to enable students to model chemical and biomolecular structures and reactions. The student will also learn how to construct scientific reports and presentations, and how to format and incorporate laboratory data appropriately into these.

Programmes

SCIE-0017	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 3 / M
SCIE-0017	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 3 / M
SCIE-0017	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 3 / M
SCIE-0017	BSc in Food Science with Business (WD_SFDSC_D)	2 / 3 / M
SCIE-0017	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 3 / M
SCIE-0017	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 3 / M

Indicative Content

- Using the ChemDraw software package to construct chemical structures, reactions & experimental diagrams
- Selecting the most effective chart type to plot laboratory data
- Plotting and using standard curves & standard addition plots
- Modelling data from a range of physical and analytical science experiments using linear regression analysis
- Evaluating correlation in laboratory data plots, including the use of residuals
- Fitting error bars to data and evaluating experimental precision
- Modelling absorbance spectra & applying derivatives to these
- Formatting data & graphs effectively for scientific reports & presentations

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Construct simple 2-D chemical structures and reactions using drawing software.
2. Apply the most effective chart type to plot laboratory data.
3. Display error bars on laboratory data points and use these to evaluate experimental precision in the data.
4. Apply linear regression to the analysis of laboratory data using spreadsheets.
5. Construct residual plots of laboratory data to evaluate correlation between data variables.
6. Construct standard addition plots and derivative curves using spreadsheets.
7. Communicate laboratory data effectively in scientific reports and presentations.

Learning and Teaching Methods

- In-class demonstrations.
- IT practicals.
- Self-directed study.
- Additional tutorial videos based on key skills learned during practicals available via e-learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	50%	3,4,5,7
<i>Practical</i>	50%	1,2,6

Assessment Criteria

- <40%:** The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to associated laboratory data analysis and presentation skills.
- 40%-49%:** The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated laboratory data analysis and presentation skills.
- 50%-59%:** As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated laboratory data analysis and presentation skills.
- 60%-69%:** In addition, the learner will demonstrate more detailed knowledge of the learning outcomes. Will display good competence with regards to associated laboratory data analysis and presentation skills.
- 70%-100%:** All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated laboratory data analysis and presentation skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	36	
Independent Learning	99	

Essential Material

- "Laboratory Data Analysis & Presentation module Moodle page." <http://moodle.wit.ie>

Supplementary Material

- Billo, E.J. _Excel for Scientists and Engineers_. Hoboken, NJ, USA: John Wiley & Sons, 2007.
- Riley, J.A. and D. Hayes. _Picture Yourself Learning Microsoft Excel 2010_. Boston,MS: Course Technology / Cengage Learning, 2010.

Requested Resources

- Room Type: Computer Lab